1.2. 使用 `git clone` 命令将仓库拉取到本地并在 `main` 文件夹里创建目录， `tree` 命令结果：

```
(base) pigeon@LAPTOP-GARN0974:~/ml-week/lumina_for_freshmen/main$ tree
.
├── Inference
│   ├── decoding
│   │   ├── decoding.py
│   │   └── registry.py
│   ├── inference_runner.py
│   ├── inferencer.py
│   └── supported.py
├── Tester
│   ├── test_runner.py
│   └── tester.py
├── Trainer
│   ├── checkpoint.py
│   └── trainer.py
├── cfg
│   ├── example_mlm.yaml
│   ├── example_mlm_inference.yaml
│   ├── example_mlm_test.yaml
│   └── example_mlm_visualize.yaml
├── check.py
├── dataset
│   ├── QA
│   │   ├── SFT
│   │   │   ├── CMRC2018
│   │   │   │   └── download.py
│   │   │   └── supported.py
│   │   ├── pretrain
│   │   │   ├── chinese_wiki
│   │   │   │   └── download.py
│   │   │   └── supported.py
│   │   └── supported.py
│   └── supported.py
├── device.py
├── docs
│   ├── Part1
│   │   ├── Task1
│   │   │   └── Task1.md
│   │   └── Task2
│   └── Part2
│       └── Task1
├── experiments.py
├── loaders
│   ├── mlm.py
│   └── qa.py
├── logger
│   └── logger.py
├── metrics
│   ├── core
│   │   ├── base.py
```

```
│   │   └── collection.py
│   ├── implementations
│   │   ├── character_f1.py
│   │   ├── exact_match.py
│   │   └── token_accuracy.py
│   ├── output
│   │   ├── plotter.py
│   │   └── writer.py
│   ├── process
│   │   └── inputs.py
│   └── registry.py
├── models
│   ├── Transformer
│   │   ├── embedding
│   │   │   ├── embedding.py
│   │   │   └── supported.py
│   │   ├── head
│   │   │   └── heads.py
│   │   ├── losses
│   │   │   ├── losses.py
│   │   │   └── registry.py
│   │   ├── modeling
│   │   │   ├── modeling.py
│   │   │   └── registry.py
│   │   ├── modules
│   │   │   ├── modules.py
│   │   │   └── supported.py
│   │   ├── optimizer
│   │   │   ├── optimizer.py
│   │   │   └── registry.py
│   │   ├── scheduler
│   │   │   ├── registry.py
│   │   │   └── scheduler.py
│   │   ├── supported.py
│   │   └── tokenizer
│   │       ├── registry.py
│   │       └── tokenizer.py
│   └── supported.py
├── parse
│   ├── bootstrap.py
│   ├── context.py
│   ├── errors.py
│   ├── parser.py
│   ├── path.py
│   ├── schema
│   │   ├── builder.py
│   │   ├── compile.py
│   │   ├── merge.py
│   │   ├── nodes.py
│   │   └── walk.py
│   ├── specs
│   │   ├── component.py
│   │   ├── registry.py
│   │   └── validator.py
```

```
│   └── validation
│       └── pipeline.py
├── registry
│   └── base.py
├── start_inference.py
├── start_test.py
├── start_train.py
├── start_visualize.py
├── tasks
│   ├── MLM_PreTrain
│   │   ├── builder.py
│   │   ├── inference.py
│   │   ├── inference_builder.py
│   │   ├── mlm.py
│   │   ├── test_builder.py
│   │   └── visualize_builder.py
│   ├── QA_SFT
│   │   ├── builder.py
│   │   ├── inference.py
│   │   ├── inference_builder.py
│   │   ├── qa.py
│   │   ├── test_builder.py
│   │   └── visualize_builder.py
│   └── registry.py
├── tokenizer
│   └── bert_base_chinese_tokenizer
│       └── download.py
└── visualize
    ├── base.py
    ├── methods
    │   ├── attention_heatmap.py
    │   └── loss_landscape.py
    └── registry.py

47 directories, 89 files
```

3.使用 `conda create -n NLP python=3.11.14 pip -y` 创建新的虚拟环境。并使用 `pip install` 安装依赖（`torch` 下载cpu版本，走清华镜像源）

```
(base) pigeon@LAPTOP-GARN0974:~/ml-week/lumina_for_freshmen/main$ conda activate NLP
(NLP) pigeon@LAPTOP-GARN0974:~/ml-week/lumina_for_freshmen/main$ conda list
# packages in environment at /home/pigeon/miniconda3/envs/NLP:
#
# Name                    Version         Build           Channel
_libgcc_mutex             0.1             main            https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/main
_openmp_mutex             5.1             52_gnu          https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/main
bzip2                     1.0.8           h5eee18b_6      https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/main
ca-certificates           2026.9.25       h06a4308_0      https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/main
contourpy                 1.3.3           pypi_0          pypi
cycler                    0.12.1          pypi_0          pypi
filelock                  3.32.3          pypi_0          pypi
fonttools                 4.67.0          pypi_0          pypi
fsspec                    2026.7.0        pypi_0          pypi
iniconfig                 2.3.1           pypi_0          pypi
jinja2                    3.1.6           pypi_0          pypi
kiwisolver                1.5.1           pypi_0          pypi
ld_impl_linux-64          2.44            h9e0c5a2_3      https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/main
libexpat                  2.9.0           h7354ed3_0      https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/main
libffi                    3.4.8           h06d3fd0_3      https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/main
libgcc                    15.2.0          h69a1729_8      https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/main
libgcc-ng                 15.2.0          h166f726_8      https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/main
libnsl                    2.0.0           h5eee18b_0      https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/main
libstdcxx                 15.2.0          h39759b7_8      https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/main
libuuid                   1.41.5          h5eee18b_0      https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/main
libxcb                    1.17.0          h9b100fa_0      https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/main
libzlib                   1.3.2           h47b2149_0      https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/main
markupsafe                3.0.3           pypi_0          pypi
matplotlib                3.10.8          pypi_0          pypi
mpmath                    1.3.0           pypi_0          pypi
narwhals                  2.27.1          pypi_0          pypi
ncurses                   6.6             hfaaeb4e_0      https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/main
networkx                  3.6.1           pypi_0          pypi
numpy                     2.3.5           pypi_0          pypi
openssl                   3.5.9           h1b28b03_0      https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/main
packaging                 26.3            py311h06a4308_0 https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/main
pillow                    13.0.0          pypi_0          pypi
pip                       26.2.1          pyhc872135_0    https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/main
plotly                    7.0.0           pypi_0          pypi
pluggy                    1.7.0           pypi_0          pypi
pthread-stubs             0.3             h47b2149_2      https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/main
```

```
mpmath                    1.3.0           pypi_0          pypi
narwhals                  2.27.1          pypi_0          pypi
ncurses                   6.6             hfaaeb4e_0      https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/main
networkx                  3.6.1           pypi_0          pypi
numpy                     2.3.5           pypi_0          pypi
openssl                   3.5.9           h1b28b03_0      https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/main
packaging                 26.3            py311h06a4308_0 https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/main
pillow                    13.0.0          pypi_0          pypi
pip                       26.2.1          pyhc872135_0    https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/main
plotly                    7.0.0           pypi_0          pypi
pluggy                    1.7.0           pypi_0          pypi
pthread-stubs             0.3             h47b2149_2      https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/main
pygments                  2.21.0          pypi_0          pypi
pyparsing                 3.3.3           pypi_0          pypi
pytest                    9.0.2           pypi_0          pypi
python                    3.11.14         h56af48b_1      https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/main
python-dateutil           2.9.0.post0     pypi_0          pypi
pyyaml                    6.0.3           pypi_0          pypi
readline                  8.3             hc2a1206_0      https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/main
setuptools                84.0.0          py311h06a4308_0 https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/main
six                       1.17.0          pypi_0          pypi
sqlite                    3.53.4          h795bf6d_0      https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/main
sympy                     1.14.0          pypi_0          pypi
tk                        8.6.15          h54e0aa7_0      https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/main
torch                     2.9.1+cpu       pypi_0          pypi
tqdm                      4.67.1          pypi_0          pypi
typing-extensions         4.16.0          pypi_0          pypi
tzdata                    2026c           he532380_0      https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/main
wheel                     0.48.0          py311h06a4308_0 https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/main
xorg-libx11               1.8.13          h65de747_0      https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/main
xorg-libxau               1.0.12          h9b100fa_0      https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/main
xorg-libxdmcp             1.1.5           h9b100fa_0      https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/main
xorg-xorgproto            2025.1          h47b2149_0      https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/main
xz                        5.8.4           h140a991_0      https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/main
zlib                      1.3.2           h47b2149_0      https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/main
```

4.



附加：仓库实现的是一个模型训练框架，主要包含 `train,test,inference,visualize` 四个主要环节，分别以 `start_xxx` 为入口调用库，以及使用 `cfg` 加载配置、 `logger` 保存日志， `model` 实现模型框架，训练实现 `mlm` 和 `qa` 两种， `mlm` 需要给出一段挖空文本让模型填空， `qa` 需要给模型一段文本和一个问题让模型回答。